

- 2 Respiration in blowfly larvae, *Calliphora erythrocephala*, can be studied by determining the respiratory quotient (RQ).

A student used the apparatus set up as shown in Fig. 2.1 to measure oxygen uptake by blowfly larvae during respiration.

Five replicates were carried out.

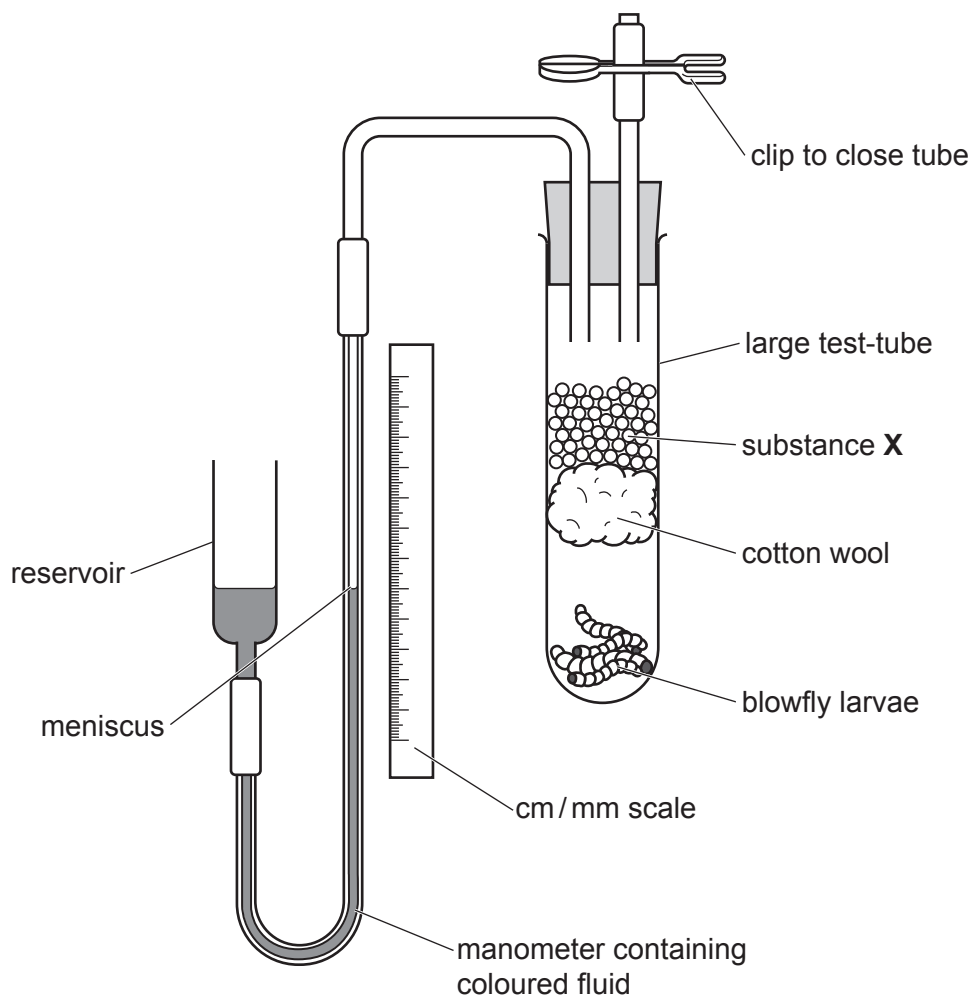


Fig. 2.1

- (a) State the role of substance X in Fig. 2.1.

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..... [1]

- (b) Predict which direction the meniscus in the manometer will move during the experiment.

Explain your answer.

direction

explanation

.....

..... [1]

- (c) To calculate the RQ, the student determined the volume of carbon dioxide produced by the blowfly larvae.

Describe how the apparatus in Fig. 2.1 was modified so that the volume of carbon dioxide produced was determined.

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..... [2]

- (d) State **two** variables that should be standardised each time the student carried out the experiments.

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..... [1]

Table 2.1 shows the results and the calculations made by the student.

Table 2.1

replicates	1	2	3	4	5
distance moved by manometer fluid in Fig. 2.1 when measuring oxygen uptake in 10 min/mm	15	48	61	56	68
volume of oxygen taken up /mm ³ min ⁻¹	1.18	3.77	4.79	4.40	
volume of carbon dioxide produced/mm ³ min ⁻¹	3.30	2.75	3.31	2.83	3.00
respiratory quotient (RQ)	2.80	0.73	0.69	0.64	

(e) The result for replicate 1, ringed in Table 2.1, is anomalous.

Suggest **two** possible causes for this anomaly other than measurement error.

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..... [2]

Fig. 2.2 shows a section of the manometer.

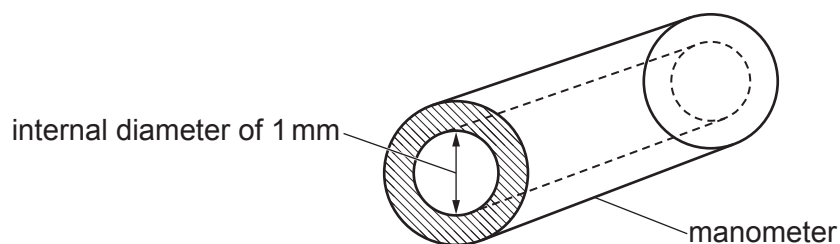


Fig. 2.2

- (f) Use the data in Table 2.1 and Fig. 2.2 to calculate the volume of oxygen taken up by the blowfly larvae in **one** minute for replicate 5.

Space for working.

volume of oxygen taken in = $\text{mm}^3 \text{min}^{-1}$ [2]

- (g) Use the answer from (f) to calculate the respiratory quotient (RQ) for replicate 5.

Space for working.

RQ = [2]

[Total: 11]